

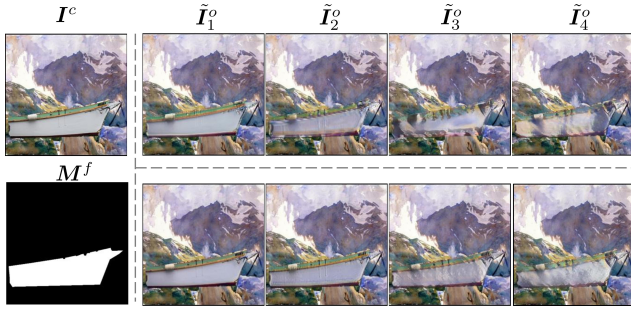


Figure 6: The top row shows the harmonized results from four stages when using the full style loss for all stages. The bottom row shows the results of our default method for comparison.

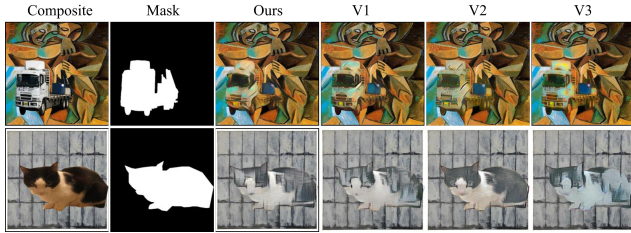


Figure 7: From left to right, we show the composite image, the composite mask, the harmonized results of our method and three ablated versions.

early-exit strategy.

4.4 Ablation Studies

Firstly, we try using the full style loss (*i.e.*, $\mathcal{L}_4^{sty}$ in Eqn. 2) for all four stages and the harmonized results from four stages are shown in the top row in Figure 6. We also show the harmonized results of our default method for comparison in the bottom row. In the top row, even using the full style loss, the outputs ($\tilde{I}_1^o$, $\tilde{I}_2^o$) from early stages can still only be harmonized up to the low style level. The content structures of outputs ($\tilde{I}_3^o$) may even be destroyed when using low-level features for high-level harmonization. The results demonstrate that low-level harmonization is simpler than high-level harmonization and high-level features are mandatory for high-level harmonization.

Secondly, we remove the losses for the first three stages and only use the loss for the final stage. As there is no supervision for the first three stages, we adopt the output from the last stage, which is referred to as V_1 . We also try some variants of our network structure. We replace the upsample decoder with bilinear upsampling, which is referred to as V_2 . We also reduce the number of fusion blocks from two to one in each stage, which is referred to as V_3 . For V_2 and V_3 , we adopt the output from the predicted exit stage. The visualization results of V_1 , V_2 , and V_3 are shown in Figure 7. The harmonized results of V_1 become worse than our full method, because the losses in the first three stages can provide useful intermediate supervision and help learn more informative

intermediate features. The harmonized results of V_2 and V_3 are also worse than our method, because the simplified network structure reduces the model capacity and degrades the harmonization ability. The above results verify the necessity of each component in our network.

5 Conclusion

In this work, we have developed a progressive harmonization network which can harmonize the composite foreground from low-level styles to high-level styles progressively. We have also designed early-exit strategy, which enables the network to exit at the proper stage. Extensive experiments on the benchmark dataset have demonstrated the superiority of our progressive harmonization network.

Acknowledgments

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